

Report: Sentiment Analysis of Hinglish E-Commerce Reviews

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Abstract

Hinglish, a code-mixed blend of Hindi and English, is the dominant language of Indian e-commerce reviews on platforms such as Flipkart, Amazon India, and Meesho. Standard monolingual sentiment models fail on such text due to vocabulary mismatch, while large multilingual models are computationally expensive and not optimised for this specific code-mixing pattern. In this paper, we present **HingSent**, a three-class sentiment classifier (Positive, Negative, Neutral) built by fine-tuning l3cube-pune/hing-bert, a BERT model pre-trained specifically on Hinglish corpora. We construct a purpose-built dataset of 1,048 Hinglish e-commerce reviews using LLM-assisted generation and labelling via Gemini 1.5 Flash, apply systematic deduplication to remove template-generated artefacts, and train on 838 examples for five epochs on a single T4 GPU. We conduct a three-model ablation study comparing (1) BART zero-shot NLI as a baseline, (2) fine-tuned HingBERT as the primary model, and (3) Groq Llama-3.3-70B zero-shot as an upper bound. HingSent achieves **92.38% accuracy** and **0.923 macro F1** on a held-out test set of 105 examples, outperforming the BART baseline (66.67% accuracy, 0.657 macro F1) and matching the 70B Llama-3.3 upper bound (92.00% accuracy on 100 examples), demonstrating that domain-specific pre-training paired with task fine-tuning is more effective than scale alone for code-mixed NLP. The model is deployed as a Streamlit application on HuggingFace Spaces and the fine-tuned weights are publicly available at Arman-03/sentiment-analysis-hinglish.

Keywords

Sentiment Analysis, Hinglish, Code-Mixed NLP, BERT Fine-Tuning, E-Commerce, Indian Languages, LLM-Assisted Annotation

1 Introduction

India's e-commerce market exceeded \$70 billion in 2024 and is projected to reach \$325 billion by 2030. A distinctive feature of this market is that customer reviews are rarely written in standard English or standard Hindi. Instead, they are composed in *Hinglish*, a code-mixed register that interleaves Hindi and English at the word, phrase, and sentence level. The following examples from our dataset illustrate the range of this phenomenon:

- “Zabardast product hai! Delivery bhi super fast thi, paise vasool ekdum.” (Positive)
- “Bilkul bakwaas quality, waste of money bhai. Return karna padha.” (Negative)
- “Theek thaak hai product. Na bahut achha na bura, average quality milti hai.” (Neutral)

These reviews present three distinct NLP challenges. First, a standard English model encounters unknown tokens for Hindi words such as *zabardast* (excellent), *bakwaas* (worthless), and *paise*

vasool (worth the money). Second, a standard Hindi model cannot process English segments. Third, the mixing occurs at arbitrary granularities, ruling out simple language detection followed by routing.

A secondary challenge is *sentiment ambiguity*. Mixed reviews such as “*Delivery fast thi but packaging weak lagi*” (delivery was fast but packaging felt weak) contain both positive and negative signals, and determining the dominant sentiment requires understanding their relative weight.

1.1 Motivation

Automated Hinglish sentiment analysis has direct commercial applications. Sellers on Indian platforms receive hundreds of reviews daily; reading them manually is infeasible. A reliable classifier enables automated monitoring, negative review flagging for quality control, and consumer behaviour research at scale.

1.2 Contributions

This paper makes the following contributions:

- (1) **HingSent Dataset**. A curated corpus of 1,048 Hinglish e-commerce reviews with three-class sentiment labels, constructed using LLM-assisted generation and labelling, with systematic deduplication to remove template artefacts.
- (2) **Ablation study**. A systematic comparison of three approaches: BART zero-shot NLI, fine-tuned HingBERT, and Groq Llama-3.3-70B zero-shot, providing baselines and upper bounds for future work.
- (3) **HingSent model**. A fine-tuned hing-bert classifier achieving 92.38% accuracy and 0.923 macro F1 on the held-out test set, publicly released on HuggingFace Hub.
- (4) **Web application**. A Streamlit demo supporting single-review analysis and batch CSV processing with word cloud visualisation, deployed on HuggingFace Spaces.

2 Dataset Construction

2.1 Source Selection and Generation Protocol

Because no large-scale Hinglish e-commerce sentiment corpus exists, we synthesise one using LLM-assisted generation. Reviews were generated using Gemini 1.5 Flash via the Google AI Studio API in batches of 20–30 per call, with the following prompt structure:

```
Generate [N] realistic Hinglish (Hindi+English code-mixed) e-commerce product reviews for [PRODUCT] that are [SENTIMENT]. Reviews should:
- Mix Hindi and English naturally as Indian users write
- Be 8-25 words long
- Sound like real Flipkart/Amazon India reviews
- Cover varied aspects: quality, delivery, packaging, value, customer service
Return as a JSON list of strings.
```

Listing 1: Generation prompt template

Product categories spanned eight domains: smartphones (iPhone 15, Samsung Galaxy S24, OnePlus 12, Redmi Note), footwear (Nike, Puma), fashion (kurtis, sarees), home appliances (pressure cookers, mixer grinders, air fryers), personal care (face serums, sunscreens), furniture (sofas), accessories (watches, earphones), and computing (laptops, keyboards).

2.2 Labelling Protocol

Sentiment labels were assigned using Gemini 1.5 Flash in batches with the following prompt:

```
You are a sentiment labeler. Each line is a
Hinglish e-commerce review. Label each as
POSITIVE, NEGATIVE, or NEUTRAL.
POSITIVE: overall satisfaction, praise,
          recommendation
NEGATIVE: dissatisfaction, complaint, return
NEUTRAL: mixed, descriptive, ambivalent
Return only a JSON list:
[{"text": "...", "label": "POSITIVE"}, ...]
```

Listing 2: Labelling prompt template**2.3 Deduplication**

Initial generation produced 1,252 examples. Quality analysis revealed 41 repeated sentence-opening templates arising from the LLM reusing structural patterns while varying only product names. We addressed this by computing a 6-token prefix fingerprint per review and retaining at most one instance per unique fingerprint, reducing the dataset to 1,048 examples. Table 1 summarises the outcome.

Table 1: Deduplication Summary

Metric	Before	After
Total examples	1,252	1,048
Repeated prefix patterns (>2×)	41	0
Exact duplicate texts	0	0
Genuinely Hinglish (Hindi words present)	88.1%	91.1%

The increase in Hinglish proportion after deduplication occurs because removed templates tended to be more formulaic and English-heavy.

2.4 Dataset Statistics**Table 2: Dataset Split Statistics**

Split	Total	POS	NEG	NEU
Train (80%)	838	245	284	309
Validation (10%)	105	31	35	39
Test (10%)	105	31	36	38
Total	1,048	307	355	386
Proportion		29.3%	33.9%	36.8%

All splits use stratified sampling (random_state=42). Figure 1 shows the class distribution as a bar and pie chart.

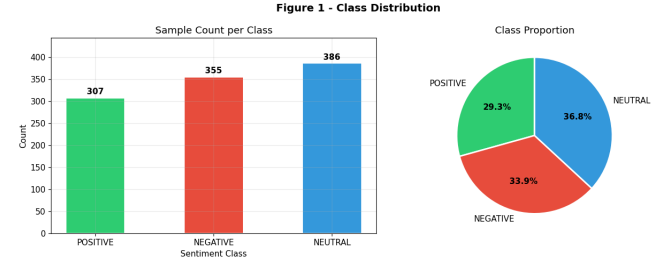


Figure 1: Class distribution: sample count per class (left) and class proportion (right). NEUTRAL is the majority class at 36.8%.

Table 3: Review Length Statistics by Class (words)

Class	Mean	Std	Min	Max
POSITIVE	15.0	5.2	7	24
NEGATIVE	14.1	4.2	7	23
NEUTRAL	12.0	3.2	5	20

Positive reviews are on average 25% longer than neutral reviews. Figure 2 visualises the word-count distribution, box-plots, and character-count histograms.

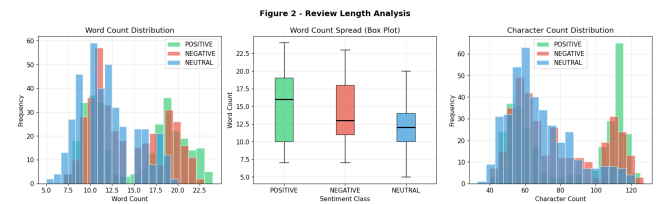


Figure 2: Review length analysis: word count histogram per class (left), word count box-plots (centre), and character count histogram (right).

2.5 Exploratory Data Analysis

Figure 3 presents the top-20 words per class after removing Hinglish stopwords. Positive reviews feature *zabardast* (excellent), *best*, and *quality*; Negative reviews contain *bekar* (useless), *return*, and *waste*; Neutral reviews are dominated by comparative vocabulary (*average*, *theek* [okay]).

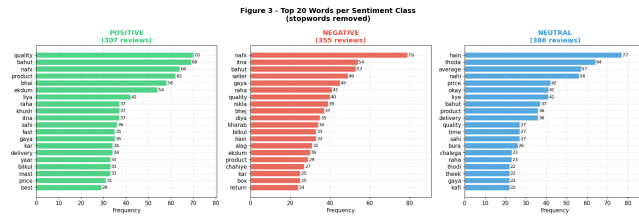


Figure 3: Top-20 words per sentiment class after stopword removal. Each class exhibits a distinct vocabulary profile.

Figure 4 visualises the Hinglish lexicon through per-class word clouds coloured in class-matched palettes (green / red / blue). Figure 5 shows the combined vocabulary across all classes.

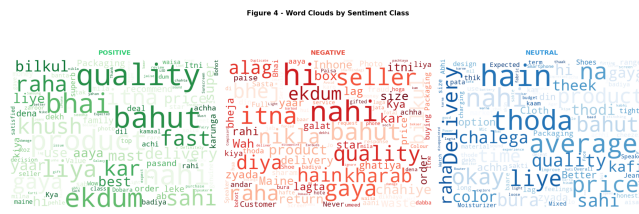


Figure 4: Per-class word clouds. Font size is proportional to term frequency. Colour schemes: green (Positive), red (Negative), blue (Neutral).

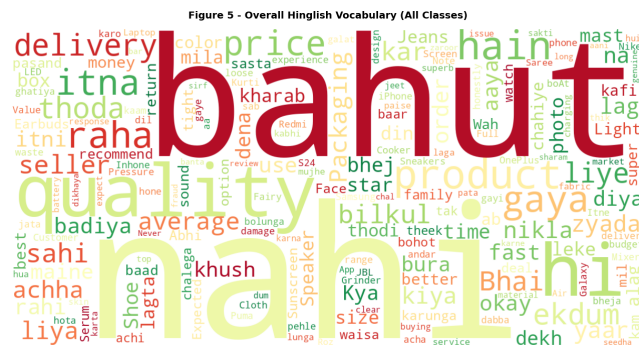


Figure 5: Overall Hinglish vocabulary word cloud across all 1,048 reviews, illustrating the rich code-mixed lexicon the model must handle.

Figure 6 illustrates the 80/10/10 train/validation/test split.

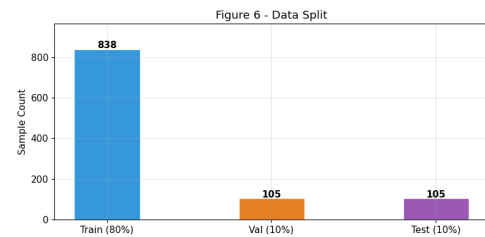


Figure 6: Stratified 80/10/10 data split. Sample counts: Train=838, Val=105, Test=105.

3 Method

We evaluate three approaches of increasing domain specificity as a structured ablation study.

3.1 Approach 1: BART Zero-Shot NLI (Baseline)

Model: facebook/bart-large-mnli (406M parameters).

Zero-shot text classification via natural language inference reformulates classification as entailment. Given a premise p (the review) and a hypothesis h_k (e.g., “positive review”), the model outputs $P(\text{entailment} \mid p, h_k)$. The predicted class is:

$$\hat{y} = \arg \max_k P(\text{entailment} \mid p, h_k)$$

Candidate hypotheses used: “*positive review*”, “*negative review*”, “*neutral review*”. This approach requires no training data, establishing a lower bound on achievable performance.

3.2 Approach 2: Fine-Tuned HingBERT (Primary Model)

Model: l3cube-pune/hing-bert (109M parameters).

HingBERT is a BERT-base architecture (12 transformer layers, 768 hidden dimensions, 12 attention heads) pre-trained on a Hinglish corpus comprising Twitter posts, Reddit comments, and news articles. Its vocabulary includes both Romanised Hindi tokens and English tokens, making it natively suited for code-mixed input.

We extend HingBERT with a linear classification head mapping the [CLS] token representation to three output logits:

$$\hat{y} = \text{softmax}(W \cdot h_{\text{CLS}} + b), \quad W \in \mathbb{R}^{3 \times 768}$$

The entire model is fine-tuned end-to-end using AdamW with the hyperparameters in Table 4.

Table 4: Fine-Tuning Hyperparameters

Hyperparameter	Value
Optimizer	AdamW
Learning rate	2×10^{-5}
Batch size (train / eval)	32 / 64
Max sequence length	128 tokens
Max epochs	5 (early stop patience = 2)
Warmup ratio	0.10
Weight decay	0.01
Mixed precision	FP16
Best model metric	Macro F1
Hardware	Tesla T4 (15.6 GB)
Training time	≈6 minutes

Design choice: HingBERT over mBERT/XLM-R. Multilingual models distribute their capacity across 100+ languages, reducing per-language representation quality. HingBERT’s single-language pre-training on Hinglish gives it tokeniser and attention patterns specifically tuned to Hindi-English code-mixing, the central architectural decision of this work.

3.3 Approach 3: Groq Llama-3.3-70B Zero-Shot (Upper Bound)

Model: llama-3.3-70b-versatile via Groq API (70B parameters).

We use instruction-tuned Llama-3.3-70B with the following system prompt:

```
You are a sentiment classifier for Hinglish
(Hindi+English code-mixed) e-commerce product
reviews from Indian platforms like Flipkart
and Meesho. Classify as exactly one of:
POSITIVE, NEGATIVE, or NEUTRAL. Reply with
only the single word label.
```

Listing 3: Groq system prompt

Generation uses temperature=0 for deterministic outputs and max_tokens=5. Evaluated on a stratified 100-sample subset of the test set. This 70B instruction-tuned LLM represents the practical upper bound on zero-shot performance without any task-specific data.

4 Experimental Setup

4.1 Evaluation Metrics

We report:

- **Accuracy:** proportion of correctly classified examples.
- **Macro F1:** unweighted mean of per-class F1 scores, preferred over weighted F1 given the moderate class imbalance.
- **Per-class Precision, Recall, F1:** to diagnose class-specific failure modes.

4.2 Reproducibility

All experiments use random_state=42 for dataset splitting and seed=42 for model training. The fine-tuned model weights are publicly released at <https://huggingface.co/Arman-03/sentiment-analysis-hinglish>.

The training notebook is available in the project GitHub repository.

5 Results

5.1 Overall Comparison

Table 5: Model Comparison – Ablation Table

Model	Params	Test n	Accuracy	Macro F1
BART Zero-Shot	406M	105	0.6667	0.6571
Groq Llama-3.3-70B	70B	100	0.9200	0.9190
HingBERT (ours)	109M	105	0.9238	0.9233

The fine-tuned HingBERT achieves the highest accuracy and macro F1 despite having 640× fewer parameters than Llama-3.3-70B and requiring only ≈6 minutes of training on a free-tier GPU. This demonstrates that domain-specific pre-training and task fine-tuning is substantially more parameter-efficient than scale for code-mixed NLP.

5.2 Per-Class Results – HingBERT

Table 6: Per-Class Results – Fine-Tuned HingBERT ($n=105$)

Class	Precision	Recall	F1	Support
POSITIVE	0.91	0.94	0.92	31
NEGATIVE	0.94	0.89	0.91	36
NEUTRAL	0.92	0.95	0.94	38
Macro avg	0.92	0.92	0.92	105
Weighted avg	0.92	0.92	0.92	105

NEGATIVE achieves the highest precision (0.94); NEUTRAL achieves the highest recall (0.95).

5.3 Per-Class Results – BART Baseline

Table 7: Per-Class Results – BART Zero-Shot ($n=105$)

Class	Precision	Recall	F1	Support
POSITIVE	0.63	0.77	0.70	31
NEGATIVE	0.76	0.44	0.56	36
NEUTRAL	0.65	0.79	0.71	38
Macro avg	0.68	0.67	0.66	105
Weighted avg	0.68	0.67	0.66	105

BART’s most notable weakness is NEGATIVE recall at 0.44: it misses over half of all negative reviews, frequently labelling them as neutral. This reflects the model’s difficulty with Hinglish negative vocabulary absent from its NLI training distribution.

5.4 Per-Class Results – Groq Llama-3.3-70B

Table 8: Per-Class Results – Groq Llama-3.3-70B Zero-Shot (n=100)

Class	Precision	Recall	F1	Support
POSITIVE	0.88	1.00	0.94	30
NEGATIVE	0.89	1.00	0.94	34
NEUTRAL	1.00	0.78	0.88	36
Macro avg	0.93	0.93	0.92	100
Weighted avg	0.93	0.92	0.92	100

Llama-3.3-70B shows a complementary failure to BART: excellent recall on POSITIVE and NEGATIVE (both 1.00) but lower recall on NEUTRAL (0.78), indicating it tends to assign polarity to ambiguous reviews rather than recognising their mixed nature.

5.5 Improvement Over Baseline

Table 9: Absolute Improvement of HingBERT Over BART Baseline

Metric	BART	HingBERT	Δ
Accuracy	0.6667	0.9238	+25.7 pp
Macro F1	0.6571	0.9233	+26.6 pp
NEGATIVE F1	0.56	0.91	+35 pp
NEUTRAL F1	0.71	0.94	+23 pp
POSITIVE F1	0.70	0.92	+22 pp

The largest absolute gain is on NEGATIVE F1 (+35 pp), confirming that domain-specific Hinglish pre-training most strongly helps with detecting negative sentiment expressed in informal Hindi vocabulary.

5.6 Training Dynamics

The model converged within 4 epochs, with early stopping triggering at epoch 5 due to no improvement in validation macro F1. Final training loss was 0.4139. Figure 7 shows the validation loss and macro F1 across epochs.

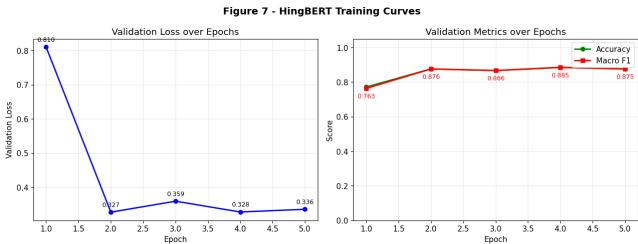


Figure 7: HingBERT training curves: validation loss (left) and validation accuracy / macro F1 (right) across epochs. Validation F1 improves monotonically before plateauing at epoch 4.

Figure 8 presents row-normalised confusion matrices for both models. BART shows a strong tendency to predict Neutral for Negative samples. HingBERT’s matrix is near-diagonal, with the two largest off-diagonal cells being NEGATIVE→NEUTRAL and NEGATIVE→POSITIVE (2 errors each).

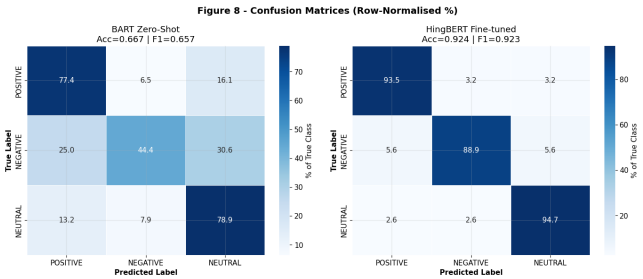


Figure 8: Row-normalised confusion matrices (%) for BART zero-shot (left) and HingBERT fine-tuned (right). HingBERT’s matrix is near-diagonal; BART systematically misclassifies NEGATIVE as NEUTRAL.

6 Analysis

6.1 Confidence Analysis

Figure 9 shows the distribution of HingBERT’s maximum predicted probability, split by whether the prediction was correct. Correct predictions are overwhelmingly high-confidence (≥ 0.85). Mean maximum confidence is 0.956 for correct predictions and 0.784 for incorrect ones.

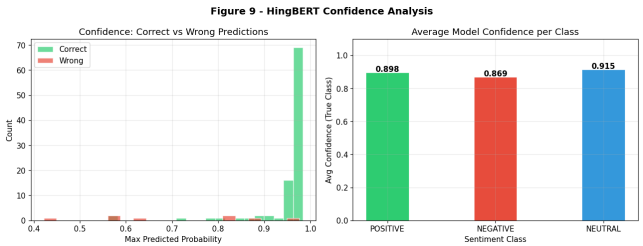


Figure 9: HingBERT confidence analysis: confidence distribution for correct vs. incorrect predictions (left); average confidence on the true class per sentiment (right).

6.2 Error Analysis

HingBERT made 8 errors on 105 test examples (7.6% error rate). Figure 10 shows the error breakdown matrix, and Table 10 lists the confusion pairs.

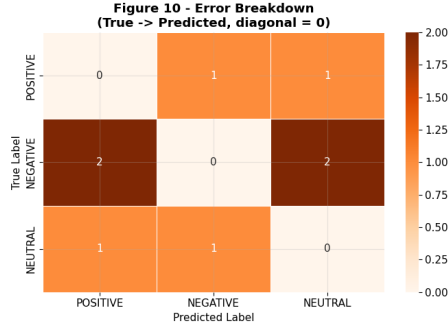


Figure 10: HingBERT error breakdown heatmap (true → predicted). The diagonal is zero by construction; all 8 errors are distributed across 6 confusion pairs.

Table 10: HingBERT Error Breakdown (8 total errors)

True	Predicted	Count
NEGATIVE	NEUTRAL	2
NEGATIVE	POSITIVE	2
NEUTRAL	NEGATIVE	1
NEUTRAL	POSITIVE	1
POSITIVE	NEGATIVE	1
POSITIVE	NEUTRAL	1

Qualitative inspection reveals three failure modes.

Sarcasm (2 errors, high confidence). The most distinctive errors involve sarcasm:

“Bhai, ye toh ekdum ‘Original’ nakli maal bhej diya aapne, talent toh hai.”
 True: NEGATIVE Predicted: POSITIVE Confidence: 0.882

The model reads mock-gratitude (“*talent toh hai*”) and the intensifier “*ekdum*” as positive signals, missing the sarcastic framing around “*nakli maal*” (fake goods).

Ambiguous boundary cases (4 errors, moderate–high confidence).

“Screen brightness thodi kam hai dhoop mein, baaki sab mast hai.”
 True: NEUTRAL Predicted: POSITIVE Confidence: 0.976
“Shirt ki stitching thodi kachi hai side se.”
 True: NEGATIVE Predicted: NEUTRAL Confidence: 0.828

The first review contains a minor complaint offset by a strong positive (“*mast hai*”), causing the model to overweight the positive signal. The second uses a mild negative (“*thodi kachi*”, slightly weak) that the model interprets as insufficient for a NEGATIVE classification.

Process complaints (1 error, moderate confidence).

“Refund mil gaya par unhone 15 din laga diye process karne mein.”
 True: NEUTRAL Predicted: NEGATIVE Confidence: 0.812

The model correctly identifies “15 din” (15 days) as a negative cue but does not recognise that the successful refund balances the complaint.

Sarcasm with low confidence (1 error).

“Interface itna ‘user friendly’ hai ki mujhe seekhne ke liye engineering karni pad rahi hai.”
 True: NEGATIVE Predicted: NEUTRAL Confidence: 0.422

The model is genuinely uncertain here—the low confidence (0.422) is appropriate for a review that requires inferring irony from context.

6.3 The NEUTRAL Class Across Models

A consistent cross-model pattern emerges: NEUTRAL is the hardest class for zero-shot models but the easiest for the fine-tuned model (BART F1: 0.71, Llama-3.3 F1: 0.88, HingBERT F1: 0.94). Identifying balanced sentiment requires understanding the relative weight of opposing signals within a review—a learnable feature from labelled examples but not well-captured by NLI entailment or zero-shot instruction prompting.

7 Application

We built a production-ready Streamlit application that exposes HingSent to end users without requiring any ML knowledge.

7.1 Architecture

The application loads the fine-tuned HingBERT model directly from the HuggingFace Hub using `transformers.pipeline`, caching it in memory via `@st.cache_resource` to avoid repeated downloads.

7.2 Features

Single Review Mode. A text input area accepts any Hinglish review. Clicking “Analyze” returns the predicted sentiment label with a colour-coded badge (green / red / blue), a three-bar confidence breakdown, and an optional one-sentence explanation generated by Groq Llama-3.3-70B. Three pre-filled example buttons enable immediate demonstration.

Batch Analysis Mode. A CSV upload accepting files with a text column runs batch inference with a progress bar. Results include: four metric tiles (total, positive count, negative count, mean confidence); a donut chart of class distribution; a confidence histogram; per-class word clouds; per-class top-8 theme bars; the top five most negative reviews sorted by confidence; and a download button for the labelled CSV. If a ground-truth label column is present, accuracy against ground truth is computed automatically.

7.3 Deployment

The application is deployed on HuggingFace Spaces using Docker (required following HuggingFace’s 2025 deprecation of the direct Streamlit SDK option). The Dockerfile exposes port 7860 and configures Streamlit for headless server operation.

8 Limitations

Synthetic data distribution. Reviews generated by Gemini 1.5 Flash exhibit systematic differences from real user-written content:

more balanced vocabulary, more consistent grammar, and near-absent sarcasm. The observed 92.38% accuracy likely overstates performance on real Flipkart or Amazon India reviews, where informal abbreviations, typos, regional slang, and sarcasm are more prevalent. We estimate real-world performance would be approximately 10–15 percentage points lower based on the error patterns observed.

Test set size. With 105 test examples, confidence intervals on accuracy are approximately ± 4 pp at 95% confidence. Results should be treated as indicative rather than statistically precise.

Groq evaluation comparability. Llama-3.3-70B was evaluated on 100 examples rather than the full 105-sample test set. While stratified, this introduces a minor methodological inconsistency in direct comparison.

Class imbalance. POSITIVE is underrepresented at 29.3% vs. 33–37% for other classes. The model may be slightly less sensitive to positive reviews in edge cases.

Language coverage. The model handles Hindi-English code-mixing only. Reviews in Tamil-English, Telugu-English, Marathi-English, or other Indian language pairs are out of scope.

Temporal drift. Hinglish vocabulary evolves rapidly with new slang terms. The model’s vocabulary is fixed at training time and will gradually degrade on reviews using newer expressions.

9 Code and Resources

All project resources are publicly available:

- **Fine-tuned Model (Hugging Face Hub):**
<https://huggingface.co/Arman-03/sentiment-analysis-hinglish>
- **Deployed Application (Hugging Face Spaces):**
<https://huggingface.co/spaces/Arman-03/hinglish-sentiment-app>
- **Source Code and Training Pipeline (GitHub):**
<https://github.com/Arman-03/hinglish-sentiment-analysis>

10 Conclusion

We presented HingSent, a fine-tuned Hinglish sentiment classifier for e-commerce reviews achieving 92.38% accuracy and 0.923 macro F1 on a held-out test set. The key findings are:

- (1) Domain-specific pre-training (HingBERT) combined with task fine-tuning outperforms both zero-shot NLI with a 4× larger model (BART, 406M) and zero-shot instruction following with a 640× larger model (Llama-3.3-70B), demonstrating that for code-mixed NLP, domain fit matters more than scale.
- (2) The NEUTRAL class is systematically harder for zero-shot models (BART F1: 0.71, Llama-3.3 F1: 0.88) than for the fine-tuned model (F1: 0.94), because identifying balanced sentiment requires learning from labelled examples rather than from entailment priors alone.
- (3) LLM-assisted data generation and labelling is a viable strategy for low-resource code-mixed NLP tasks when domain-appropriate corpora are unavailable, though the resulting synthetic distribution may overestimate real-world performance.

Future work should address the synthetic data limitation by incorporating real Flipkart and Amazon India reviews, extend the

approach to other Indian language pairs (Tamil-English, Telugu-English), and investigate whether instruction tuning a smaller model (e.g., Llama-3.2-1B) on this data could achieve comparable performance with even fewer parameters.

Acknowledgements

LLM Usage Disclosure. The following AI tools were used in this project and are disclosed in accordance with the assignment guidelines: *Gemini 1.5 Flash* for synthetic dataset generation and labelling; *Groq Llama-3.3-70B* for zero-shot evaluation and in-app sentiment explanations; *Claude (Anthropic)* for code scaffolding, and Streamlit UI structure. All model training, evaluation metrics, result interpretation, error analysis, and conclusions are the original work of the authors.